Coduri Bose–Chaudhuri–Hocquenghem (BCH)

Implementare Python

Model de canal

$$c \in \mathbb{F}_2^n, \quad r = c + e$$

$$wt(e) \leq t$$

$$C \subseteq \mathbb{F}_2^n \text{ cod liniar}$$

Obiectiv:

$$\forall e,\ wt(e) \leq t \Rightarrow c \text{ recuperabil}$$

Corpuri finite

$$|K| = p^m$$

$$K \cong \mathbb{F}_p[x]/(P(x))$$

$$\deg P = m, \quad P \text{ ireductibil}$$

Bază:

$$\{1, \alpha, \alpha^2, \ldots, \alpha^{m-1}\}$$

$$\alpha = x \mod P(x)$$

Cazul binar

$$GF(2^m)$$

$$|GF(2^m)^\times| = 2^m - 1$$

$$\exists\alpha : \langle\alpha\rangle = GF(2^m)^\times$$

$$a \neq 0 \Rightarrow a = \alpha^i$$

Coduri ciclice

Lungime:

$$n = 2^m - 1$$

Identificare:

$$(c_0, \ldots, c_{n-1}) \leftrightarrow c(x) = \sum c_i x^i$$

$$C = \langle g(x) \rangle, \quad g(x) \mid (x^n - 1)$$

Rădăcini impuse

Alegem:

$$g(\alpha^b) = g(\alpha^{b+1}) = \cdots = g(\alpha^{b+2t}) = 0$$

Cod BCH restrâns:

$$b = 1$$

Corectează:

$$t \text{ erori}$$

Frobenius

$$f(x) \in \mathbb{F}_2[x], \;\; f(\beta) = 0$$

$$\Rightarrow f(\beta^2) = 0$$

Orbita:

$$\beta, \beta^2, \beta^{2^2}, \ldots$$

Cosete ciclotomice

$$n = 2^m - 1$$

$$C_i = \{i, 2i, 4i, 8i, \dots\} \pmod{n}$$

Partiție:

$$\{0, 1, \dots, n-1\} = \bigsqcup C_i$$

Polinom minimal

Pentru α^i:

$$M_i(x) = \prod_{j \in C_i} (x - \alpha^j)$$

$$M_i(x) \in \mathbb{F}_2[x]$$

Ireductibil.

Generator BCH

$$g(x) = \mathrm{lcm}(M_1, M_2, \ldots, M_{2t})$$

Evitare duplicare:

$$M_i = M_j \text{ dacă } i, j \in C_k$$

Limita BCH

Dacă:

$$g(\alpha^b) = \cdots = g(\alpha^{b+\delta-2}) = 0$$

atunci:

$$d_{\min} \geq \delta$$

Pentru BCH restrâns:

$$d_{\min} \geq 2t + 1$$

Codare

Mesaj:

$$m(x)$$

$$c(x) = x^r m(x) + \rho(x)$$

unde:

$$\rho(x) = x^r m(x) \mod g(x)$$

$$r = \deg g$$

Sindroame

$$S_k = r(\alpha^k), \quad k = 1, \ldots, 2t$$

Dacă:

$$S_k = 0 \ \forall k \Rightarrow \text{fără erori}$$

Berlekamp–Massey

Căutăm:

$$\Lambda(x) = 1 + \lambda_1 x + \cdots + \lambda_L x^L$$

$$S_k + \lambda_1 S_{k-1} + \cdots + \lambda_L S_{k-L} = 0$$

$$L \leq t$$

Căutare Chien

Poziții erori:

$$\Lambda(\alpha^{-i}) = 0$$

Pentru:

$$i = 0, \ldots, n - 1$$

Implementare (schematic)

$$GF(2^m): \text{ tabele exp/log}$$

$$\text{cosete } \Rightarrow M_i(x)$$

$$g(x) = \prod M_i(x)$$

Flux:

$$\text{encode} \to \text{syndroame} \to \Lambda(x) \to \text{Chien}$$

Complexitate

$$GF: \; O(2^m)$$

$$\text{Sindroame}: \; O(nt)$$

$$BM: \; O(t^2)$$

$$Chien: \; O(nt)$$

Concluzie

Algebră $\Rightarrow$ corecție eficientă

BCH: parametri controlați

Implementabil rapid